

MP491 2018/2019: Tutorial Sheet 1

1-dim ODEs

For each of the following equations you should locate the equilibrium points and investigate their stability properties for varied values of the parameter a . You should then draw the summary/bifurcation diagram, including arrows, for the problem indicating the qualitative behaviour of the non-equilibrium solutions for all values of a , including at the bifurcation point $a = a^*$. Note any bifurcations that occur and determine which of the following terms can be used to describe the nature of the bifurcation you find: **transcritical**, **subcritical**, **supercritical**, **pitchfork**, **saddle-node/fold**. Discuss the relevance of the three theorems found in the course notes to each of these problems, i.e. **show that the conditions hold**.

1.

$$\frac{dx}{dt} = ax + 3x^2$$

2.

$$\frac{dx}{dt} = ax + 2x^3$$

3.

$$\frac{dx}{dt} = a + x^2$$

4.

$$\frac{dx}{dt} = ax - 4x^3$$

5.

$$\frac{dx}{dt} = 1 - ax^2$$